

Lección 15

M. Siddikov

May 4, 2026

Plan de trabajo

Teorema de Noether

—Formulación (repaso)

*relación con simetrías

**Resolución de problemas

Formulación lagrangiana de partícula relativista

► Nuestro Syllabus:

FIS-310:

UNIDAD 4. Formulación Lagrangiana de la Partícula Relativista

4.1. Lagrangiano de la Partícula Relativista

4.2. Ecuaciones de Lagrange para Partículas Relativista

4.3. Aplicaciones

Certamen:

—Miércoles 04/05/2026

Summary of the last lecture

- ① The action of the free relativistic particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}, \quad L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

► Conserved quantities:

- energy E (consequence of invariance w.r.t. $t \rightarrow t + t_0$)
- momentum $\mathbf{p} = \partial L / \partial \mathbf{v} = m\mathbf{v} / \sqrt{1 - v^2/c^2}$ (consequence of translational invariance, $\mathbf{r} \rightarrow \mathbf{r} + \mathbf{r}_0$)
- angular momentum (projection to any axis) $M_i = \partial L / \partial \dot{\phi}_i$ (consequence of invariance w.r.t. rotation around any axis)

- ② Interaction: Add the interaction term to the action:

$$S_0 \rightarrow S = S_0 + S_{\text{int}}$$

- Depends on the nature of interaction
- Must be Lorentz- and rotation covariant,
- Must be translation-invariant
- Should NOT include long-distance interaction $U(\mathbf{r}_i - \mathbf{r}_j)$

Relativistic motion in central field

Example: Relativistic Kepler Problem

–Lagrangian:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - U(r) = -mc^2 \sqrt{1 - \frac{\dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)}{c^2}} + \frac{\alpha}{r}, \quad \alpha > 0$$

–generalized coordinates: r, θ, ϕ

–generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = \gamma m \dot{r}, \quad \gamma \equiv \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \gamma m r^2 \dot{\theta},$$

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \gamma m r^2 \sin^2 \theta \dot{\phi} = \text{const}$$

–Explicit Euler-Lagrange equations are extremely complicated, so instead will use integrals of motion to find the trajectory

–**Symmetries** & integrals of motion:

$$\phi \rightarrow \phi + \phi_0 \Rightarrow p_\phi \equiv M_z = \text{const}, \quad t \rightarrow t + t_0 \Rightarrow E = \text{const}$$

$$E = p_r \cdot \dot{r} + p_\theta \cdot \dot{\theta} + p_\phi \cdot \dot{\phi} - L = mc^2 \gamma - \frac{\alpha}{r}, \quad \alpha > 0$$

–In spherical coordinates we don't see other symmetries, but ...

Relativistic Kepler Problem

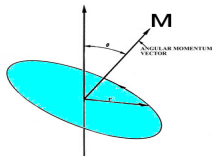
- In Cartesian coordinates we know that potential is invariant w.r.t. rotation around *any* axis



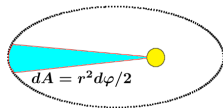
- All components of $\mathbf{M}$ are conserved, $\mathbf{M}$ keeps its direction
- In spherical coordinates the orientation of basis vectors $\mathbf{e}_\theta, \mathbf{e}_\phi$ changes, so we don't see this property (actually there is a hidden symmetry which mixes θ, ϕ in nontrivial way)

Consequences of $\vec{M} = \text{const}$:

- Particle always moves in a plane orthogonal to $\vec{M}$, i.e. vectors $(\vec{r}(t), \vec{v}(t))$ always remain in this plane, for $\forall t$.
- We may choose plane $\theta = \pi/2$ as a plane of motion, like in nonrelativistic case



$$M \equiv \gamma m r^2 \dot{\phi} = \frac{m r^2 \dot{\phi}}{\sqrt{1 - v^2/c^2}} = \text{const}$$



Second law of Kepler is NOT valid

- Due to $\gamma \neq \text{const}$, see that $r^2 d\phi \neq \text{const}$ for relativistic particle in general.
- It works ONLY for special case when particle has $v = \text{const}$ (energy conservation: this is possible only along circular orbit with $r = \text{const}$)

Relativistic Kepler Problem

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\mathbf{M}$)

$$E = m\gamma c^2 + U(r), \quad \gamma = \left[1 - \frac{v^2}{c^2}\right]^{-1/2} = \left[1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2}\right]^{-1/2}$$

$$M_z = m\gamma r^2 \dot{\phi} = \text{const} \quad \Rightarrow \quad M_z^2 \left(1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2}\right) = m^2 r^4 \dot{\phi}^2$$

$$\dot{\phi} = M_z \sqrt{\frac{1 - \dot{r}^2/c^2}{m^2 r^4 + M_z^2 r^2/c^2}}, \quad \Rightarrow \quad \gamma = \frac{M_z}{mr^2 \dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r} \quad (1)$$

In nonrelativistic case ($c \rightarrow \infty$) this result yields

$$E_{\text{nonrel}} = E - mc^2 = \frac{m\dot{r}^2}{2} + \frac{M_z^2}{2mr^2} - \frac{\alpha}{r} + \mathcal{O}\left(\frac{1}{c^2}\right)$$

and we interpreted the last 2 terms as an effective potential.

–The region $r = 0$ was “protected” by centrifugal forces: for finite energy E , we could not achieve $r = 0$.

Relativistic Kepler Problem

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\mathbf{M}$)

$$E = m\gamma c^2 + U(r), \quad \gamma = \left[1 - \frac{v^2}{c^2}\right]^{-1/2} = \left[1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2}\right]^{-1/2}$$

$$M_z = m\gamma r^2 \dot{\phi} = \text{const} \quad \Rightarrow \quad M_z^2 \left(1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2}\right) = m^2 r^4 \dot{\phi}^2$$

$$\dot{\phi} = M_z \sqrt{\frac{1 - \dot{r}^2/c^2}{m^2 r^4 + M_z^2 r^2/c^2}}, \quad \Rightarrow \quad \gamma = \frac{M_z}{mr^2 \dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r} \quad (1)$$

In **nonrelativistic case** ($c \rightarrow \infty$) this result yields

$$E_{\text{nonrel}} = E - mc^2 = \frac{m\dot{r}^2}{2} + \frac{M_z^2}{2mr^2} - \frac{\alpha}{r} + \mathcal{O}\left(\frac{1}{c^2}\right)$$

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Relativistic Kepler Problem

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r} \quad (1)$$

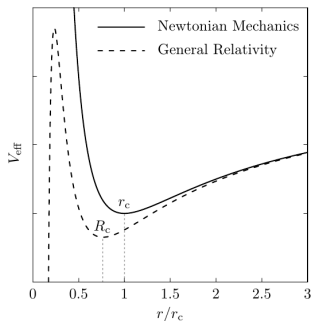
– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} \left(E + \frac{\alpha}{r}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right) \quad (2)$$

– Can interpret the last term as “potential” and use it for qualitative interpretation, however there is some peculiarity:

- * “Potential” depends on true energy E
- * Should consider mc^2 as “energy” (see left-hand side)
- * **Large- c expansion near $E \approx mc^2$**

$$V_{\text{eff}} \approx -\frac{\alpha}{r} + \frac{M_z^2}{2mr^2} - \frac{\alpha M_z^2}{mc^2} \frac{1}{r^3} + \frac{\alpha^2 M_z^2}{2m^2 c^4} \frac{1}{r^4}$$



Relativistic Kepler Problem

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r} \quad (1)$$

– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} \left(E + \frac{\alpha}{r}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right) \quad (2)$$

Solution of EOMs:

$$\frac{dr}{dt} = f(r, E) = \sqrt{c^2 - m^2 c^6 \left(E + \frac{\alpha}{r}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}$$
$$\int^r \frac{dr}{f(r, E)} = t - t_0 \quad (3)$$

The integral over r is expressed in terms of elliptic functions.

Relativistic Kepler Problem

Solution of EOMs:

$$\frac{dr}{dt} = f(r, E) = \sqrt{c^2 - m^2 c^6 \left(E + \frac{\alpha}{r}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}, \quad \gamma = \dots = \frac{1}{mc^2} \left(E + \frac{\alpha}{r}\right)$$

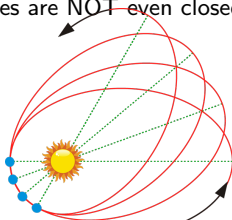
–Equation for trajectory $r(\phi)$:

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{M_z c^2}{r^2 f(r, E)} \left(E + \frac{\alpha}{r}\right) \equiv g(r, E)$$

$$\Rightarrow \phi - \phi_0 = \int dr g(r, E)$$

*Trajectories are NOT conical sections, the 1st and 3rd Kepler's laws are NOT valid.

*Trajectories are NOT even closed (precession for Mercury orbit: 14.3"/century).



Generalization: Relativistic motion in central field

(Generalization from Kepler Problem)

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - U(r)$$

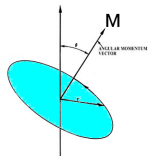
Energy conservation:

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} + U(r) = mc^2 \gamma + U(r) = \text{const}, \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Angular momentum conservation:

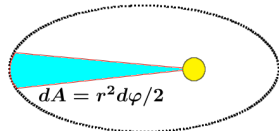
$$\mathbf{M} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\gamma \mathbf{v} = \text{const}$$

–Conservation of the plane of motion, can choose as $\theta = \pi/2$



$$M \equiv \gamma m r^2 \dot{\phi} = \frac{m r^2 \dot{\phi}}{\sqrt{1 - v^2/c^2}} = \text{const}$$

2nd law of Kepler (area law) is NOT valid
–If $\gamma \neq \text{const} \Rightarrow dA = r^2 d\phi/2 \neq \text{const } dt$



Relativistic motion in central field

Motion in plane $\theta = \pi/2$ orthogonal to $\mathbf{M}$, solution of conditions $E = \text{const}$, $M_z = \text{const}$

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} + U(r) \quad \dot{\phi} = M_z \sqrt{\frac{1 - \dot{r}^2/c^2}{m^2 r^4 + M_z^2 r^2/c^2}}$$

In **nonrelativistic case** ($c \rightarrow \infty$) this result yields

$$E_{\text{nonrel}} = E - mc^2 \approx \frac{m\dot{r}^2}{2} + \frac{M_z^2}{2mr^2} + U(r) + \mathcal{O}\left(\frac{1}{c^2}\right)$$

In general case we may rewrite $E = \text{const}$ and get

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)$$

Looks like nonrelativistic equation for 1D motion, but:

- "Energy" (left-hand side) is always $mc^2/2$
- The last term in right-hand side gives "effective" potential, depends on true energy E and angular momentum M_z

$$\text{lim}_{c \rightarrow \infty} : \quad \varepsilon = E - mc^2 = \frac{m\dot{r}^2}{2} + V_{\text{eff}}^{(\text{nonrel})}(r)$$

$$V_{\text{eff}}^{(\text{nonrel})} \approx U(r) + \frac{M_z^2}{2mr^2} - \frac{M^2 (\varepsilon - U(r))}{m^2 r^2 c^2} + \frac{3(\varepsilon - U(r))^2}{2mc^2} + \dots$$

Trajectory: Relativistic motion in central field

Solution of EOMs:

$$t - t_0 = \int \frac{dr}{\sqrt{c^2 - m^2 c^6 (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}}, \quad \phi(t) = \int dt \frac{M_z}{mr^2} \underbrace{\frac{mc^2}{(E - U(r))}}_{\gamma}$$

–Equation for trajectory $r(\phi)$:

$$\phi - \phi_0 = \int dr \frac{M_z c^2 (E - U(r))}{r^2 \sqrt{c^2 - m^2 c^6 (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}}$$

Example - motion in the potential $U = kr^2/2$

A **relativistic** particle of mass m moves in the spherically symmetric potential $U(r) = kr^2/2$ of 3D harmonic oscillator. Find explicitly the relativistic $\mathcal{O}(v^2/c^2)$ corrections to classical trajectory. Analyze if the trajectory will remain closed

Reminder - NR results:

$$L = \frac{mv^2}{2} - \frac{\kappa(x^2 + y^2 + z^2)}{2}$$

—get 3 decoupled equations:

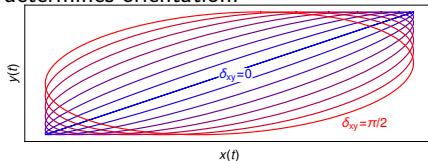
$$m\ddot{x}_a = -\kappa x_a, \quad a = 1, 2, 3$$

$$x_a = A_a \cos(\omega t + \varphi_a), \quad a = 1, 2, 3$$

—Potential is central $\Rightarrow$ motion is always in plane transverse to $\mathbf{M}$

*If we choose basis vector $\hat{\mathbf{z}}$ in direction $\mathbf{M}$, then $x_3 \equiv 0$ ($A_3 = 0$)

—equation for ellipse; $\delta_{xy} = \varphi_x - \varphi_y$ determines orientation:



* Different colors = different δ_{xy}

* $r = 0$ corresponds to center, NOT focus of the ellipse like in Kepler problem

Motion in the potential $U = \kappa r^2/2$

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} + \frac{\kappa r^2}{2} \approx mc^2 + \frac{m\dot{r}^2}{2} + \frac{M_z^2}{2mr^2} + \frac{\kappa r^2}{2} + \mathcal{O}(v^2/c^2)$$

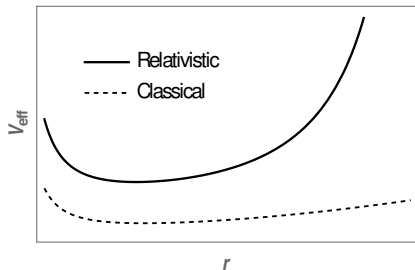
– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} \left(1 + \left(\frac{M_z}{mrc} \right)^2 \right) / \left(\frac{\kappa r^2}{2} \right)^2$$

Can interpret the last term as “potential” and use it for qualitative interpretation, however there is some peculiarity:

- * “Potential” depends on true energy E
- * Should consider mc^2 as “energy” (see left-hand side)
- * Large- c expansion near $E \approx mc^2$

$$V_{\text{eff}} \approx \frac{\kappa r^2}{2} + \frac{M_z^2}{2mr^2} + \frac{\kappa (4M_z^2 + 3m\kappa r^4)}{8m^2 c^2}$$



Motion in the potential $U = kr^2/2$

Solution of EOMs:

$$\frac{dr}{dt} = f(r, E) = \sqrt{c^2 - m^2 c^6 \left(E - \frac{\kappa r^2}{2}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}$$

$$\int^r \frac{dr}{f(r, E)} = t - t_0 \quad (3)$$

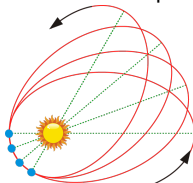
The integral over r is expressed in terms of elliptic functions.

–Equation for trajectory $r(\phi)$:

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{M_z}{mc^2 f(r, E)} \left(E - \frac{\kappa r^2}{2}\right) \equiv g(r, E)$$

$$\Rightarrow \phi - \phi_0 = \int dr g(r, E)$$

*Trajectories are NOT ellipses, NOT even closed.



Corrections to NR trajectory

–jump back to Cartesian coordinates (easier analysis)

Effective potential:

$$\tilde{V}_{\text{eff}} \approx \frac{\kappa r^2}{2} + \frac{\kappa (4M_z^2 + 3m\kappa r^4)}{8m^2 c^2} \approx \frac{\kappa (x^2 + y^2 + z^2)}{2} + \frac{\lambda}{4} (x^2 + y^2 + z^2)^2$$

–grey-colored term does not effect EOM

Equations of motion:

$$m\ddot{x} = -\kappa x + \lambda x (x^2 + y^2 + z^2),$$

$$m\ddot{y} = -\kappa y + \lambda y (x^2 + y^2 + z^2),$$

$$m\ddot{z} = -\kappa z + \lambda z (x^2 + y^2 + z^2),$$

Solution of EOMs (perturbative approach):

$$x \approx A_x \cos(\omega t + \varphi_x) + \lambda x_1(t) + \lambda^2 x_2(t) + \dots$$

$$y \approx A_y \cos(\omega t + \varphi_y) + \lambda y_1(t) + \lambda^2 y_2(t) + \dots$$

$$z \approx 0$$

$$\ddot{x}_1 + \omega_0^2 x_1 \approx \lambda (A_x^3 \cos^3(\omega t + \varphi_x) + A_x A_y^2 \cos(\omega t + \varphi_x) \cos^2(\omega t + \varphi_y)) = f_x(t)$$

$$\ddot{y}_1 + \omega_0^2 y_1 \approx \lambda (A_y A_x^2 \cos^2(\omega t + \varphi_x) \cos(\omega t + \varphi_y) + A_y^3 \cos^3(\omega t + \varphi_y)) = f_y(t)$$

Corrections to NR trajectory

$$\ddot{x}_1 + \omega_0^2 x_1 \approx \lambda (A_x^3 \cos^3(\omega t + \varphi_x) + A_x A_y^2 \cos(\omega t + \varphi_x) \cos^2(\omega t + \varphi_y)) = f_x(t)$$

$$\ddot{y}_1 + \omega_0^2 y_1 \approx \lambda (A_y A_x^2 \cos(\omega t + \varphi_y) \cos^2(\omega t + \varphi_x) + A_y^3 \cos^3(\omega t + \varphi_y)) = f_y(t)$$

Solution:

$$x_1(t) = \int^t d\tau G(t - \tau) f_x(\tau), \quad y_1(t) = \int^t d\tau G(t - \tau) f_y(\tau),$$

where G is Green function

$$G(\xi) = \int \frac{d\omega}{2\pi} \frac{e^{i\omega\xi}/m}{(-\omega^2 + \omega_0^2)} = \frac{\theta(\xi)}{m\omega_0} \sin(\omega_0\xi)$$

Similarly can restore term-by-term higher order corrections

Control question

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc} \right)^2 \right)$$

Assume that a relativistic particle moves in the field of attractive central potential

$$U(r) = -\frac{\alpha}{r^n}, \quad \alpha, n = \text{const}, \quad \alpha > 0$$

- We have seen that in nonrelativistic case for $n < 2$, the particle with nonzero angular momentum $M_z = m\rho v_\infty \neq 0$ will never fall on center (reach point $r = 0$).
- Analyze for what values of n the same statement will be true for relativistic particle (assume $E > mc^2$, $M_z = m\rho v_\infty \neq 0$).

Help:

- Try to relate the fall on center with existence of the “turning point” ($\dot{r} = 0$)
- Pay attention to behaviour of different terms in the limit $r \rightarrow 0$

Lagrangian in (external) electro-
magnetic field

Motion in external vector (=electromagnetic) field

Landau: Minimal action for a free particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

postulate that for real trajectory $S = \min$.

- Now we will consider motion in external 4-vector field $A_\mu(x)$
 - We'll see that this reproduces equations for electromagnetism. Our goal is just to learn how to write lagrangian covariantly, so we will not discuss any experimental aspects/foundations.
- We assume that interaction might be introduced by additional interaction term S_{int} in our action
- Result should be covariant (the same in all frames)

Motion in external EM field

▷ Let's assume

$$S_{\text{int}} = -q \int dx_{\mu} A^{\mu}(x) = q \int d\tau (u \cdot A),$$
$$u_{\mu} \equiv \frac{dx_{\mu}}{d\tau}; \quad u \cdot A \equiv \eta_{\mu\nu} u^{\mu} A^{\nu}$$

If we use notation $A^{\mu} = (\phi, \mathbf{A})$, then

$$S_{\text{int}} = -q \int d\tau \frac{\phi - \mathbf{A} \cdot \mathbf{v}}{\sqrt{1 - v^2/c^2}} = -q \int dt (\phi - \mathbf{A} \cdot \mathbf{v})$$
$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q(\phi - \mathbf{A} \cdot \mathbf{v})$$

Write out the Equations of Motion for such lagrangian and demonstrate that it agrees with Lorentz formula.

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Motion in external EM field

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q(\phi - \mathbf{A} \cdot \mathbf{v})$$

► Generalized momentum:

$$P^a = \frac{\partial L}{\partial v^a} = \underbrace{\frac{mv^a}{\sqrt{1 - \frac{v^2}{c^2}}}}_{p^a} + q A^a$$

► p^a -momentum of free particle; $P^a \neq p^a$

► Equation of motion:

$$\frac{dP^a}{dt} \equiv \frac{dp^a}{dt} + q \frac{dA^a(x, t)}{dt} = \frac{\partial L}{\partial r^a} = -q \nabla_a \phi + q \nabla_a (\mathbf{A} \cdot \mathbf{v})$$

Motion in external EM field

$$\frac{dP^a}{dt} \equiv \frac{dp^a}{dt} + q \frac{dA^a(x, t)}{dt} = \frac{\partial L}{\partial r^a} = -q \nabla_a \phi + q \nabla_a (\mathbf{A} \cdot \mathbf{v})$$

► Claim:

$$\nabla (\mathbf{A} \cdot \mathbf{v}) \equiv \mathbf{v} \times [\nabla \times \mathbf{A}] + (\mathbf{v} \cdot \nabla) \mathbf{A}$$

$$(\mathbf{a} \times [\mathbf{b} \times \mathbf{c}] = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}), \text{ so } \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) = \mathbf{a} \times [\mathbf{b} \times \mathbf{c}] + \mathbf{c}(\mathbf{a} \cdot \mathbf{b}))$$

► Claim:

$$\frac{d\mathbf{A}(x, t)}{dt} \equiv \frac{\partial \mathbf{A}(x, t)}{\partial t} + \frac{dx^b}{dt} \frac{\partial \mathbf{A}(x, t)}{\partial x^b} = \left[\frac{\partial \mathbf{A}(x, t)}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{A} \right]$$

so

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}$$

► Magnetic field does not change kinetic energy:

$$\frac{dE_{\text{kin}}}{dt} = \mathbf{v} \cdot \frac{d\mathbf{p}}{dt} = q \mathbf{E} \cdot \mathbf{v}$$

Motion in external electromagnetic field

► Explicit derivation of energy E from the Noether's theorem:

$$E = \sum_a \frac{\partial L}{\partial v_a} v_a - L = \text{const}$$

$$\mathbf{P} = \frac{\partial L}{\partial \mathbf{v}} \equiv \mathbf{p} + q \mathbf{A}$$

$$\begin{aligned} E &= \mathbf{v} \cdot \mathbf{P} - L = \frac{mv^2}{\sqrt{1 - v^2/c^2}} + \cancel{q \mathbf{A} \cdot \mathbf{v}} + mc^2 \sqrt{1 - \frac{v^2}{c^2}} + q \left(\phi - \cancel{\mathbf{A} \cdot \mathbf{v}} \right) \\ &= \frac{mc^2}{\sqrt{1 - v^2/c^2}} + q \phi \end{aligned}$$

—Magnetic potential $\mathbf{A}$ does not contribute to energy, in agreement with our expectations

Motion in external EM field (covariant)

[Sec. 7.10 in *Goldstein*]

► So far we demonstrated that the action

$$S = S_0 + S_{\text{int}}$$
$$S_0 = -mc^2 \int d\tau, \quad S_{\text{int}} = q \int d\tau (u \cdot A)$$

correctly reproduced Lorentz force, yet we found that the expression is NOT Lorentz-covariant.

— *Reason*: We explicitly separated time t and coordinates x

► **Our goal**: get covariant form of equations of motion

— *Method*:

*We'll assume the trajectory is given by $x^\mu(\tau)$ and NOT separating explicitly coordinates and time in lab-frame.

*We'll analyze change of action δS in case of variation of the trajectory

$$x^\mu(\tau) \rightarrow x^\mu(\tau) + \delta x^\mu(\tau)$$

and after that request $\delta S / \delta x^\mu = 0$.

Motion in external EM field (covariant)

► Variation of integrand of S_0 :

$$\begin{aligned}d\tau &\equiv \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} \\ \delta d\tau &= \frac{\eta_{\mu\nu} dx^\mu d\delta x^\nu}{\sqrt{\eta_{\alpha\beta} dx^\alpha dx^\beta}} = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} d\delta x^\nu = \eta_{\mu\nu} u^\mu d\delta x^\nu \equiv u_\nu d\delta x^\nu\end{aligned}$$

► Variation of integrand of S_{int}

$$\delta(dx_\mu A^\mu(x)) = (d\delta x_\mu) A^\mu(x) + (dx_\mu) \frac{\partial A^\mu(x)}{\partial x^\nu} \delta x^\nu$$

► For the terms $\int d\tau f_\mu(\dots) d\delta x^\mu$ we use integration by parts

$$f_\mu(x) d\delta x^\mu \equiv d(f_\mu(x) \delta x^\mu) - df_\mu(\dots) \delta x^\mu = d(f_\mu(x) \delta x^\mu) - \frac{\partial f_\mu(\dots)}{\partial x^\nu} dx^\nu \delta x^\mu$$

$$\int_{\tau_1}^{\tau_2} d\tau f_\mu(\dots) d\delta x^\mu = f_\mu(x) \delta x^\mu \Big|_{\tau_1}^{\tau_2} - \int \frac{\partial f_\mu(x)}{\partial x^\nu} dx^\nu \delta x^\mu = - \int d\tau \frac{\partial f_\mu(x)}{\partial x^\nu} u^\nu \delta x^\mu$$

Motion in external EM field (covariant)

► Variation of action:

$$\begin{aligned}\delta S &= \int d\tau \delta x^\mu \left[mc^2 \frac{du^\nu}{d\tau} - q F^{\mu\alpha} u_\alpha \right] = 0, \\ F_{\mu\nu} &\equiv \partial_\mu A_\nu - \partial_\nu A_\mu \\ m \frac{du^\nu}{d\tau} &\equiv \frac{dp^\nu}{d\tau} = q F^{\mu\nu} u_\nu \quad (1)\end{aligned}$$

► $F_{\mu\nu}$ is called electromagnetic field tensor

Write out explicitly all the components of $F_{\mu\nu}$ and demonstrate that (1) is equivalent to

$$\begin{aligned}\frac{d\mathbf{p}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \\ \mathbf{E} &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}\end{aligned}$$

(assume $A^\mu = (\phi/c, \mathbf{A})$, $A_\mu = \eta_{\mu\nu} A^\nu = (\phi/c, -\mathbf{A})$)

Motion in external EM field (covariant)

$$m \frac{du^\mu}{d\tau} \equiv \frac{dp^\mu}{d\tau} = q F^{\mu\nu} u_\nu \quad (1)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$d\tau = dt \sqrt{1 - v^2/c^2}$$

$$u^0 = u_0 = \frac{c}{\sqrt{1 - v^2/c^2}}$$

$$u^j = -u_j = \frac{v^j}{\sqrt{1 - v^2/c^2}},$$

► Explicit form:

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

-factors $1/\sqrt{1 - v^2/c^2}$ appear in both parts and cancel

▷ Eq. (1) in components ($p_0 \equiv E_{\text{kin}}/c$):

$$F_{\mu\nu} = \eta_{\mu\alpha} F^{\alpha\beta} \eta_{\beta\nu} =$$

$$= \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

▷ Use $\eta_{\mu\nu}$, $\eta^{\mu\nu}$ to raise/lower indices
-signs in front of electric field change when raising/lowering indices, so be careful

$$\begin{aligned} \frac{dp_0}{dt} &= \frac{q}{c} (E_x v_x + E_y v_y + E_z v_z) = \frac{q}{c} \mathbf{E} \cdot \mathbf{v} \\ -B_y \frac{dp_x}{dt} &= q (E_x + v_y B_z - v_z B_y), \\ 0 \frac{dp_y}{dt} &= q (E_y + v_z B_x - v_x B_z), \\ \frac{dp_z}{dt} &= q (E_z + v_x B_y - v_y B_x), \end{aligned}$$

$$E_i = c F_{0i}, \quad B_i = -\frac{1}{2} \epsilon_{ijk} F^{jk}$$

in agreement with results which we got in lab-frame

Control question

– In our derivation we *assumed* that interaction term is linear in A^μ ,

$$S_{\text{int}} = -q \int dx_\mu A^\mu(x) = q \int d\tau (u \cdot A),$$
$$u_\mu \equiv \frac{dx_\mu}{d\tau}; \quad u \cdot A \equiv \eta_{\mu\nu} u^\mu A^\nu$$

What happens if we include higher-order terms, e.g.

$$S_{\text{int}}^{(\text{extra})} = \lambda_2 \int d\tau (u \cdot A)^2 + \lambda_3 \int d\tau (u \cdot A)^3$$

- *Are these terms compatible with Lorentz-covariance of the action?*
- *Why physically these terms should not be included for electromagnetism?*

Lagrangian for a system of N particles

Write out the lagrangian of a system which consists of N electric charges moving with relativistic velocities

Lagrangian for a system of N particles

► We cannot use long-range forces, contradicts relativity

– ... instead should consider as a system N particles + EM field

$$S = S_0 + S_{\text{int}} + \int d^2x \mathcal{L}_{\text{field}}$$

► Interaction always has local form. Nonlocal terms would contradict special relativity

► Should include term $\mathcal{L}_{\text{field}}$ which describes the dynamics of the field itself

– Lagrangian $\mathcal{L}_{\text{field}}$ can be constructed only from $F_{\mu\nu}$ (gauge invariant)

Courses of electromagnetism:

1) The only possible construction:

$$\mathcal{L}_{\text{field}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

2) Variation of A_μ yields extra equations for electromagnetic field (Maxwell's equations):

$$\partial_\nu F^{\mu\nu} = \mu_0 j^\mu \quad \epsilon_{\alpha\beta\mu\nu} \partial^\beta F^{\mu\nu} = 0$$

Lagrangian in (external) gravitational field

Motion in external gravitational field

Landau: Minimal action for a free particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

postulate that for real trajectory $S = \min$.

- Up to now we always assumed that the interaction might be introduced by additional interaction term S_{int}
- Adding S_{int} is not the only covariant way to introduce interaction. For example, we could replace flat metrics

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

- as $\eta_{\mu\nu} \rightarrow g_{\mu\nu}(x)$, where matrix $g_{\mu\nu}(x)$ called “metric tensor”.
- We’ll also use notation $g^{\mu\nu}(x)$ for the inverse
- Eventually this leads to General Relativity (GR, theory of gravity)
- ... complicated topic, we will only touch it to learn how to write covariantly the equations of motion. We will omit all the experimental evidence/checks
- Our discussion will be mostly introductory and based on [Chapter 7 in Goldstein]
- ... for more detailed discussion see [Landau, Vol. 2] (though we will not go that far, there is a dedicated course on this)

Motion in external gravitational field

Landau: Minimal action for a particle:

$$S = -mc^2 \int d\tau \sqrt{g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}$$

$-\tau$ could be any parameter along the particle's trajectory. If it is proper time, then $\sqrt{g_{\mu\nu} \dots} = 1$

► As usual assume the variation vanishes at the endpoints

► Einstein's convention implied (summation over dummy repetitive indices)

► Variation of particle's trajectory:

$$\begin{aligned} -\delta S / mc^2 &= \\ &= \int d\tau \frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \delta x^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau g_{\mu\nu}(x) \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{d\delta x^\nu}{d\tau} \\ &= \int d\tau \delta x^\lambda \left(\frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} - \frac{\partial}{\partial \tau} \left(g_{\mu\lambda}(x) \frac{dx^\mu}{d\tau} \right) - \frac{\partial}{\partial \tau} \left(g_{\lambda\mu}(x) \frac{dx^\mu}{d\tau} \right) \right) \\ &\quad \frac{\partial g_{\mu\lambda}(x)}{\partial \tau} \rightarrow \frac{\partial x^\nu}{\partial \tau} \frac{\partial g_{\mu\lambda}(x)}{\partial x^\nu} \end{aligned}$$

$$0 = \int d\tau \delta x^\mu \left(-2g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

Motion in external gravitational field

$$0 = \int d\tau \delta x^\mu \left(-2g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

$$g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{1}{2} \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} (\partial_\alpha g_{\mu\nu} + \partial_\nu g_{\mu\alpha} - \partial_\mu g_{\alpha\nu}) = 0$$

► "Equations of motion" (geodesics):

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0$$
$$\Gamma_{\alpha\beta}^\mu = \frac{g^{\mu\nu}}{2} \left(\frac{\partial g_{\alpha\nu}}{\partial x^\beta} + \frac{\partial g_{\nu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right)$$

► Symbols Γ are called Cristoffel's symbols

► We may rewrite

$$\frac{dp^\mu}{d\tau} = -m \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}$$

so the right-hand side might be interpreted as a covariant 4-force which is proportional to mass m (recall equality of gravitational and inertial masses)

Hamiltonian formalism

Hamiltonian formulation (intro)

Since usually the Hamiltonian (canonical) formulation is given in undergraduate courses, we need to try to recall as much as possible in order to proceed

...

– Please participate

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
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Control question 2

Frequently students say: “Hamiltonian is energy!”

- Is this statement correct/accurate?
- How can we reformulate this statement more carefully ?

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Control question 3

Is Hamiltonian approach applicable in presence of non-potential (e.g. dissipative, velocity-dependent magnetic) forces?

Control question 4

Assume that You have a system of 3 objects: the Sun, the Earth and the Moon which interact gravitationally with each other.

—How many hamiltonians do You need to describe this system?